

ByeolPass – Final Report

1. Introduction

ByeolPass is a Java-based password security tool designed to generate and audit passwords through both a command-line interface and a graphical user interface. The project combines practical password security features with a modular software design.

2. Application Functionalities

2.1 Password Generation

The application allows users to generate one or multiple passwords with customizable options such as:

- password length
- number of generated passwords
- exclusion of special characters
- exclusion of ambiguous characters
- letters-only mode
- digits-only mode
- exclusion of uppercase letters
- exclusion of lowercase letters
- raw output mode
- file output support

2.2 Password Audit

The application also supports password auditing. This feature evaluates password strength using criteria such as:

- minimum length
- lowercase letters
- uppercase letters
- digits
- special characters
- repeated patterns
- sequential patterns
- keyboard patterns
- common weak passwords

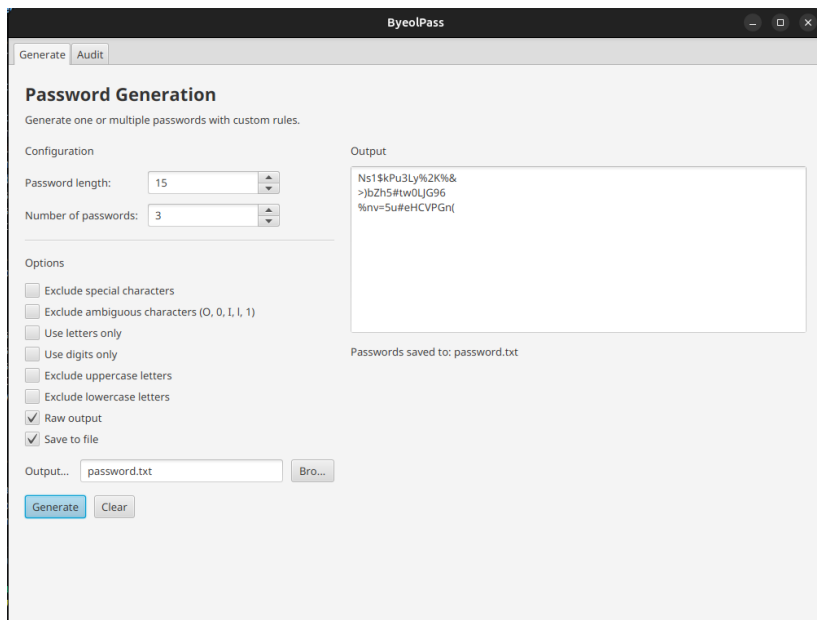
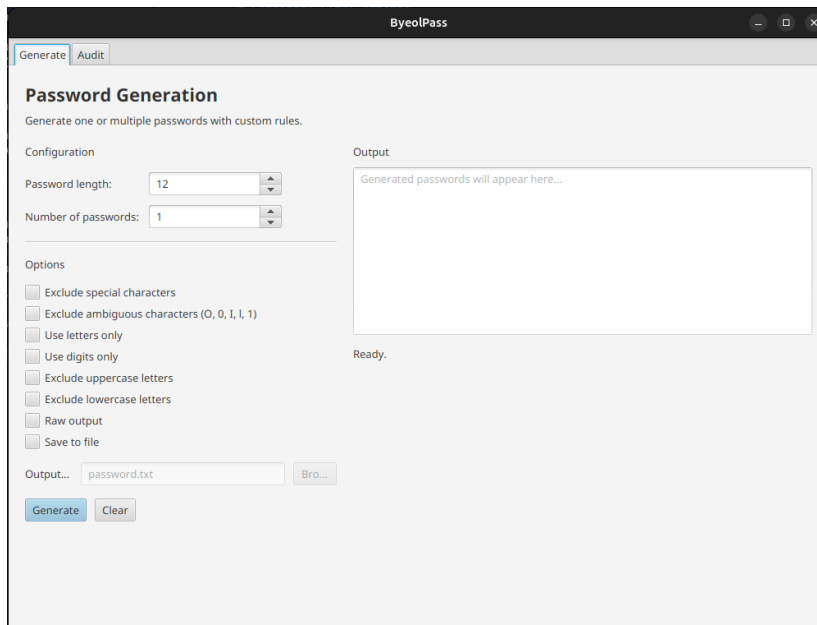
The audit output includes a score, a strength level, detected issues, and suggestions.

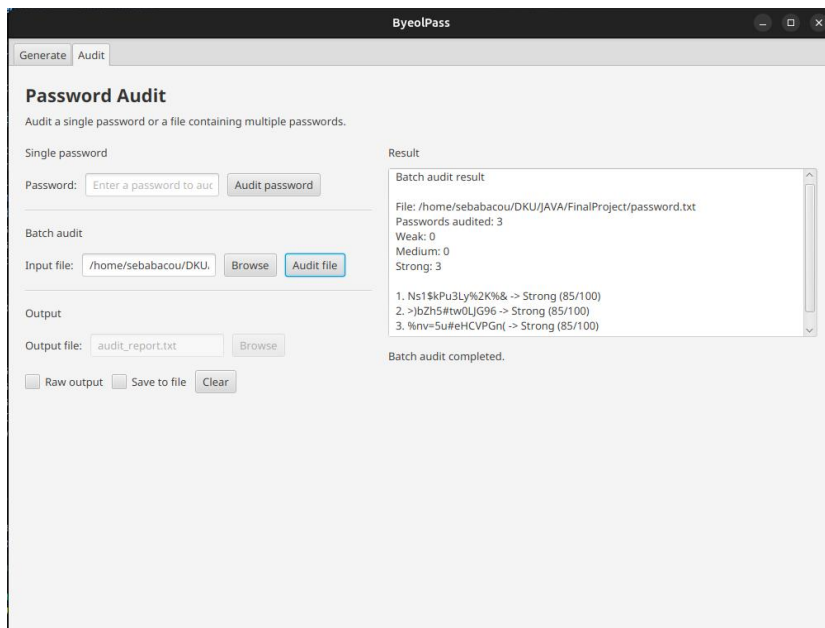
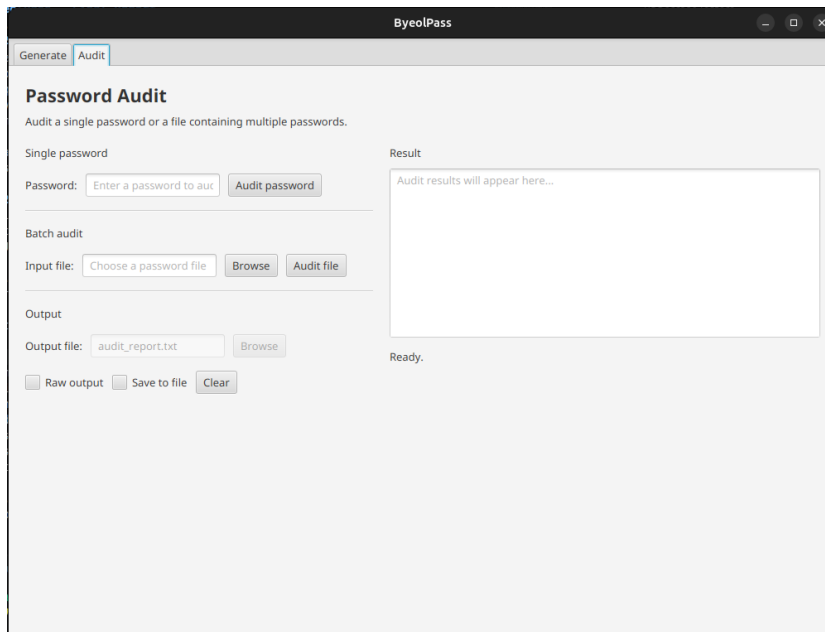
2.3 Interfaces

ByeolPass includes:

- a command-line interface for direct usage and testing
- a JavaFX graphical interface for easier interaction

3. Application Screenshots





4. Design and Implementation

The project follows a modular architecture separating business logic from interfaces.

Main components:

- PasswordGenerator
- PasswordPolicy
- PasswordAuditor
- CLI command classes

- JavaFX GUI views

This structure improves readability, maintenance, and future scalability.

5. Additional Notes

This project allowed me to work on Java object-oriented design, CLI command parsing, exception handling, file export, and GUI development. The backend logic was developed first and later connected to the graphical interface.

6. Conclusion

ByeolPass is a complete password security application that combines password generation and password auditing in a modular Java project. It can be extended further with more advanced password analysis and management features.

7. GitHub Repository

GitHub link: [ByolPass](#)